Module 7a slides: Bioavailability and Bioequivalence (worked examples)

Module 7a - Bioavailability and Bioequivalence · Lecture slides, worked examples · 30 Sep 2026

Name _____ Date _____

Practice Problem

The bioavailability of an investigational drug was studied in 6 volunteers. Each volunteer received either a single oral tablet containing 250 mg of the drug or a single IV bolus injection containing 100 mg of the drug. The average AUC of the oral tablet was 101.5 (mcg/mL)hr and the average AUC of the IV bolus injection was 73.8 (mcg/mL)hr. What is the absolute bioavailability of the drug from the tablet?

Practice Problem (follow-up)

The absolute bioavailability of a drug from an oral tablet was found to be 0.55. A patient is receiving 100 mg of the drug by IV bolus. What oral dose of the tablet would provide the same extent of absorption as the 100 mg IV bolus dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Practice Problem Her key

Answer: 0.55 (as a decimal fraction)

WHAT THE STEM GIVES

n	6 volunteers	*not needed: the AUCs are already averages*
D_{po}	250 mg	the denominator of the dose ratio
D_{IV}	100 mg	the numerator of the dose ratio
AUC_{po}	101.5 (mcg/mL)hr	the numerator of the area ratio
AUC_{IV}	73.8 (mcg/mL)hr	the denominator of the area ratio

THE WORKING, ONE LINE AT A TIME

SETUP

$$F_{abs} = \left(\frac{AUC_{po}}{AUC_{IV}}\right)\left(\frac{D_{IV}}{D_{po}}\right)$$

Absolute bioavailability compares the tablet with the IV (intravenous) bolus, so the IV AUC (area under the curve) is the denominator. The doses differ, 250 mg against 100 mg, so the dose ratio is needed, with the IV dose on top.

UNIT

Both AUCs are in (mcg/mL)hr and both doses in mg, so the units cancel

The AUC ratio is (mcg/mL)hr over (mcg/mL)hr and the dose ratio is mg over mg. Nothing is left, so F is dimensionless and the units can be dropped from the working.

ALGEBRA

$$F = \left(\frac{101.5}{73.8}\right)\left(\frac{100\ mg}{250\ mg}\right) = 1.3753 \times 0.4 = 0.5501$$

The tablet's AUC is larger than the IV AUC only because its dose was two and a half times larger. Multiplying by 100 over 250 scales the tablet's area to what a 100-mg tablet would have given, and that is 55 per cent of the IV area.

ROUND

$F = 0.55$

Her value, 0.55: 55 per cent of the oral dose is available for the body to use, against all of an IV dose. Written as 0.55 or as 55 per cent, both are accepted; the F carries into the next part, the equivalent oral dose.

> **DOES THE NUMBER MAKE SENSE?**
> A 250 mg tablet gave only 1.3753 times the area of a 100 mg IV dose; dose for dose that is 1.3753 × 0.4 = 0.55. F must lie between 0 and 1.

Write F as 0.55 or as 55%, never as .55: there should be no leading decimals.

Practice Problem (follow-up) Her key

Answer: 181.8 mg

WHAT THE STEM GIVES

F	0.55	the denominator
D_{IV}	100 mg	the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

Equivalent therapy: $AUC_{po} = AUC_{IV}$, so $F = \frac{D_{IV}}{D_{po}}$

The same therapeutic effect means the same extent of exposure, which is the same AUC (area under the curve). With the AUC ratio equal to 1, the absolute bioavailability equation leaves only the dose ratio.

ALGEBRA

$$D_{po} = \frac{D_{IV}}{F}$$

Multiply both sides by D_{po} and divide by F. Dividing the IV dose by a fraction smaller than 1 gives a larger oral dose, as expected: only 55 per cent of what is swallowed reaches the circulation, so more has to be swallowed.

ALGEBRA

$$D_{po} = \frac{100 \text{ mg}}{0.55} = 181.8 \text{ mg}$$

F is dimensionless, so milligrams divided by it stay milligrams. The oral dose is 1.82 times the IV dose, the reciprocal of 0.55.

ROUND

D_{po} = 182 mg, to be rounded to a strength that exists

Her value, 182 mg from 181.818. A tablet of 181.8 mg is not made, so the prescribed dose is rounded to a strength that is: 200 mg if 175 and 200 are the strengths available. The calculated figure is the one keyed here.

> **DOES THE NUMBER MAKE SENSE?**
>
> Only 0.55 of an oral dose reaches the blood, so the oral dose has to be larger than the 100 mg IV dose, and less than twice it because F is above 0.5: 181.8 mg.